

Code Tracing examples

Script:

```
index = 0
while index<3:
    index = index + 1
    print(index)
```

Trace:

index	Output

Script:

```
index = 0
while index<3:
    print(index)
    index = index + 1
```

Trace:

index	Output

Script:

```
a = 5
b = 7
while b<10:
    b = b + 1
    print(b-a)
```

Trace:

a	b	Output

Script:

```
a = b = 0
while b-a >= 0:
    b = b + 1
    a = a + b - 1
    print(b-a)
```

Trace:

a	b	Output

Script:

```
a = b = 0
while b-a > 0:
    b = b + 1
    a = a + b - 1
    print(b-a)
```

Trace:

a	b	Output

Script:

```
index = total = 0
while index != 5:
    index += 1
    if index > 2:
        total += index
    print(total)
```

Trace:

index	total

What is the output? _____

Script:

```
index = total = 0
while index != 5:
    index += 2
    if index > 2:
        total += index
print(total)
```

Trace:

index	total

What went wrong? _____

How could we fix it?
